



Hybrid Compressed Air Assisted Propulsion System for Cargo Ships

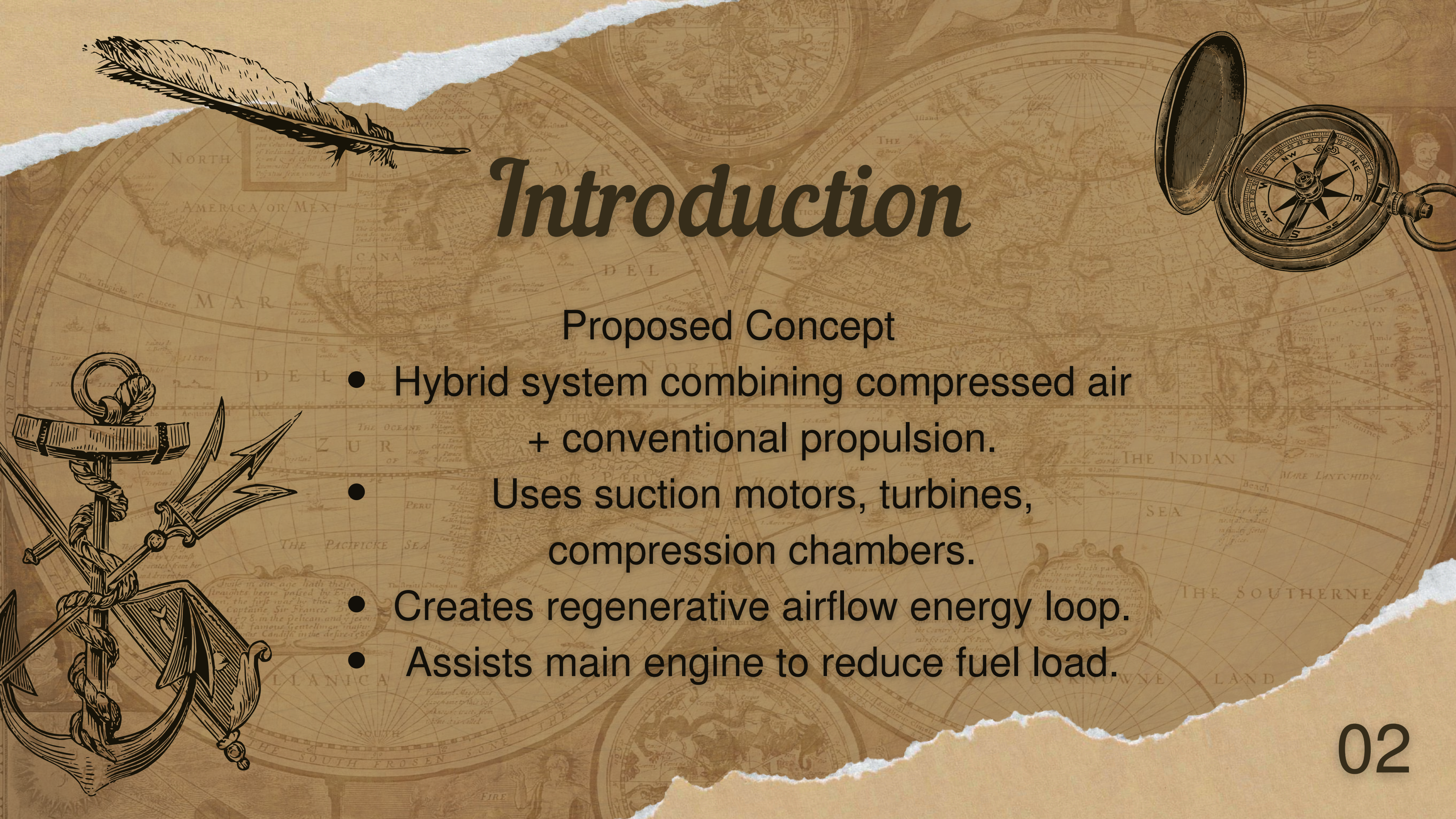
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Introduction

Background

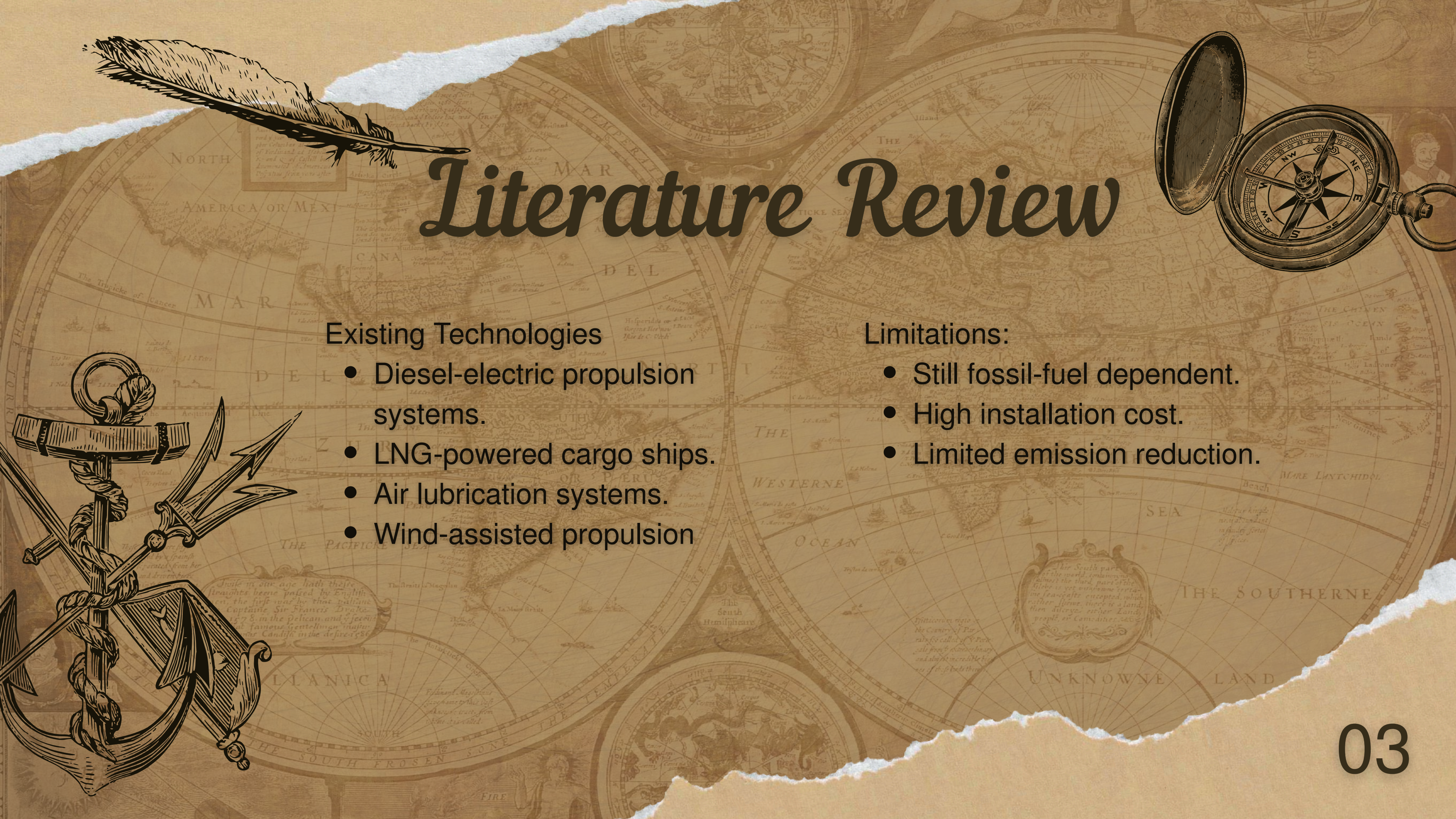
- Maritime transport handles 80% of global trade.
- Cargo ships rely heavily on diesel engines.
- High fuel consumption increases operational cost.
- Marine sector contributes significantly to CO₂ emissions.



Introduction

Proposed Concept

- Hybrid system combining compressed air + conventional propulsion.
- Uses suction motors, turbines, compression chambers.
- Creates regenerative airflow energy loop.
- Assists main engine to reduce fuel load.



Literature Review

Existing Technologies

- Diesel-electric propulsion systems.
- LNG-powered cargo ships.
- Air lubrication systems.
- Wind-assisted propulsion

Limitations:

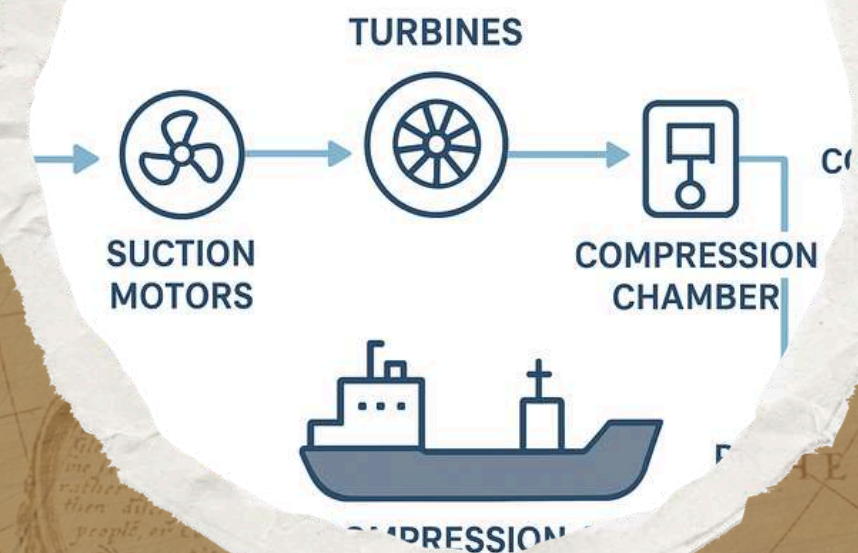
- Still fossil-fuel dependent.
- High installation cost.
- Limited emission reduction.

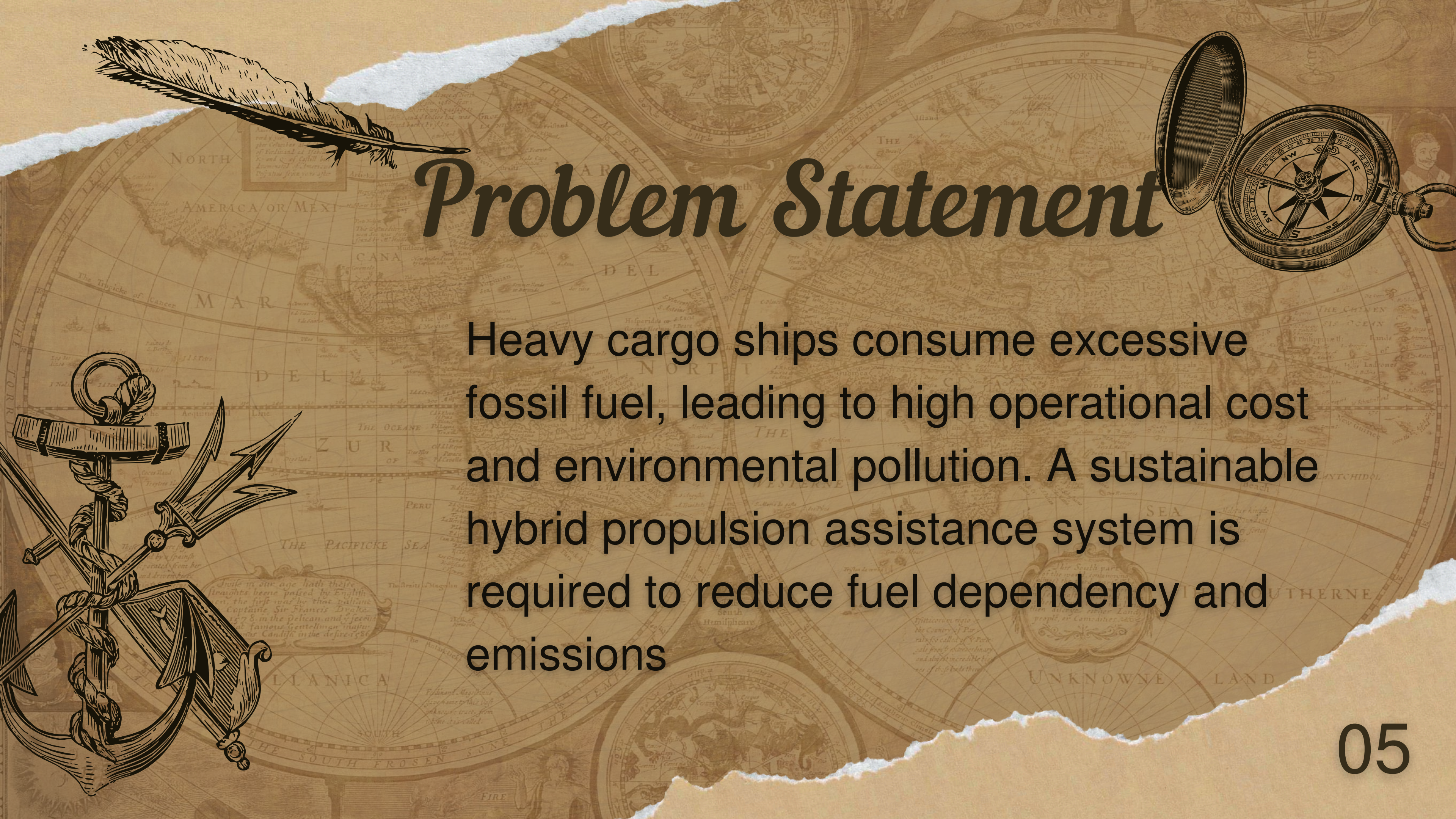
Literature Review

Research Gaps

- Lack of regenerative air-based propulsion in maritime.
- No integrated compressed air hybrid model.
- Limited research on airflow-driven auxiliary propulsion.

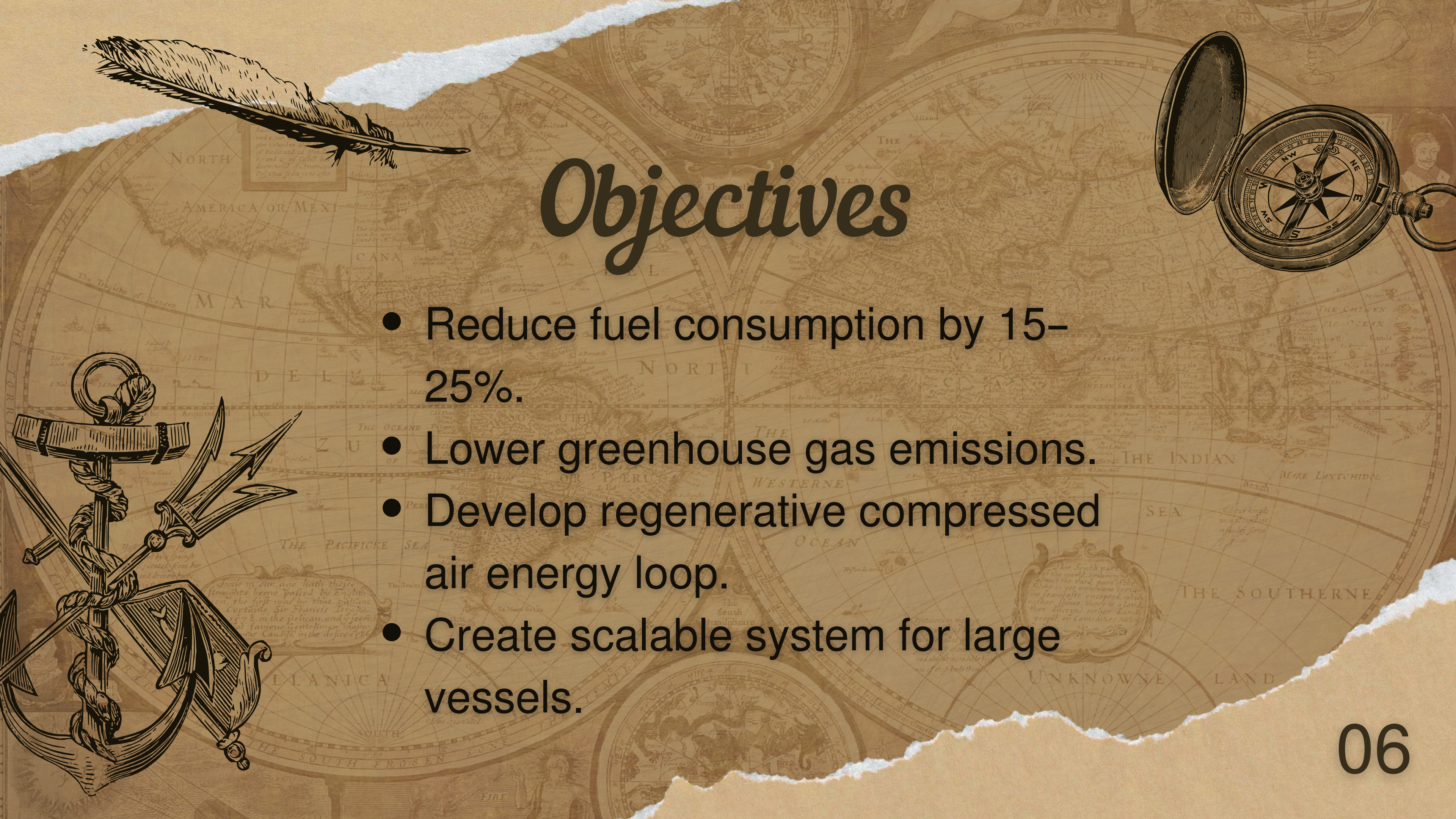
COMPRESSED AIR PROPULSION SYSTEM FOR CARGO SHIPS TO REDUCE FUEL CONSUMPTION





Problem Statement

Heavy cargo ships consume excessive fossil fuel, leading to high operational cost and environmental pollution. A sustainable hybrid propulsion assistance system is required to reduce fuel dependency and emissions



Objectives

- Reduce fuel consumption by 15–25%.
- Lower greenhouse gas emissions.
- Develop regenerative compressed air energy loop.
- Create scalable system for large vessels.

Proposed Methodology

System Architecture Components:

- Front Suction Motors
- Air Turbines
- Compression Chamber
- Secondary Turbines
- Propeller Assistance System

Proposed Methodology

Working Principle – Step 1

1. Suction motors draw ambient air.
2. Air passes through turbine.
3. Turbine generates electricity.
4. Electricity recharges battery.

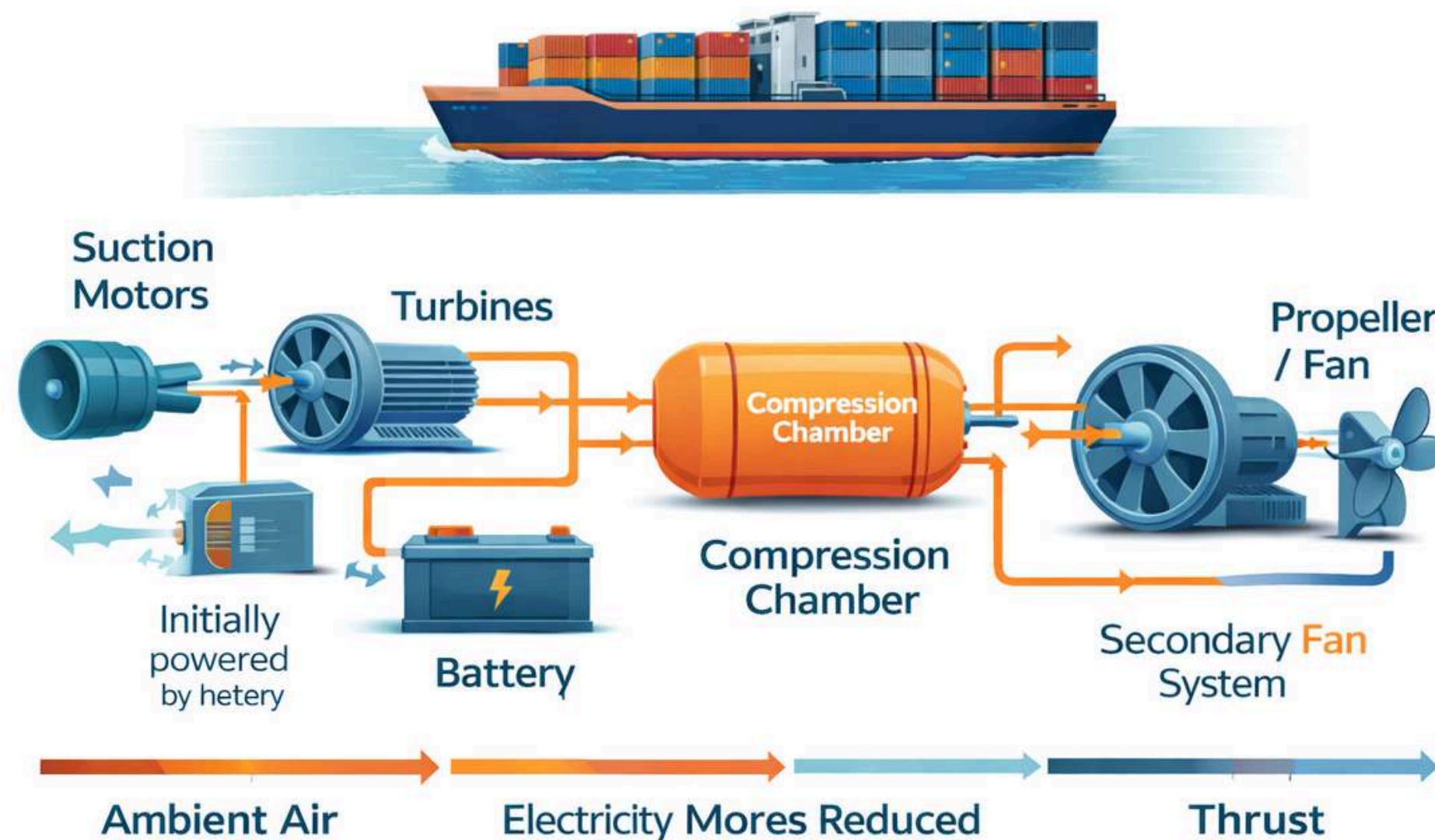
Proposed Methodology

Working Principle – Step 2

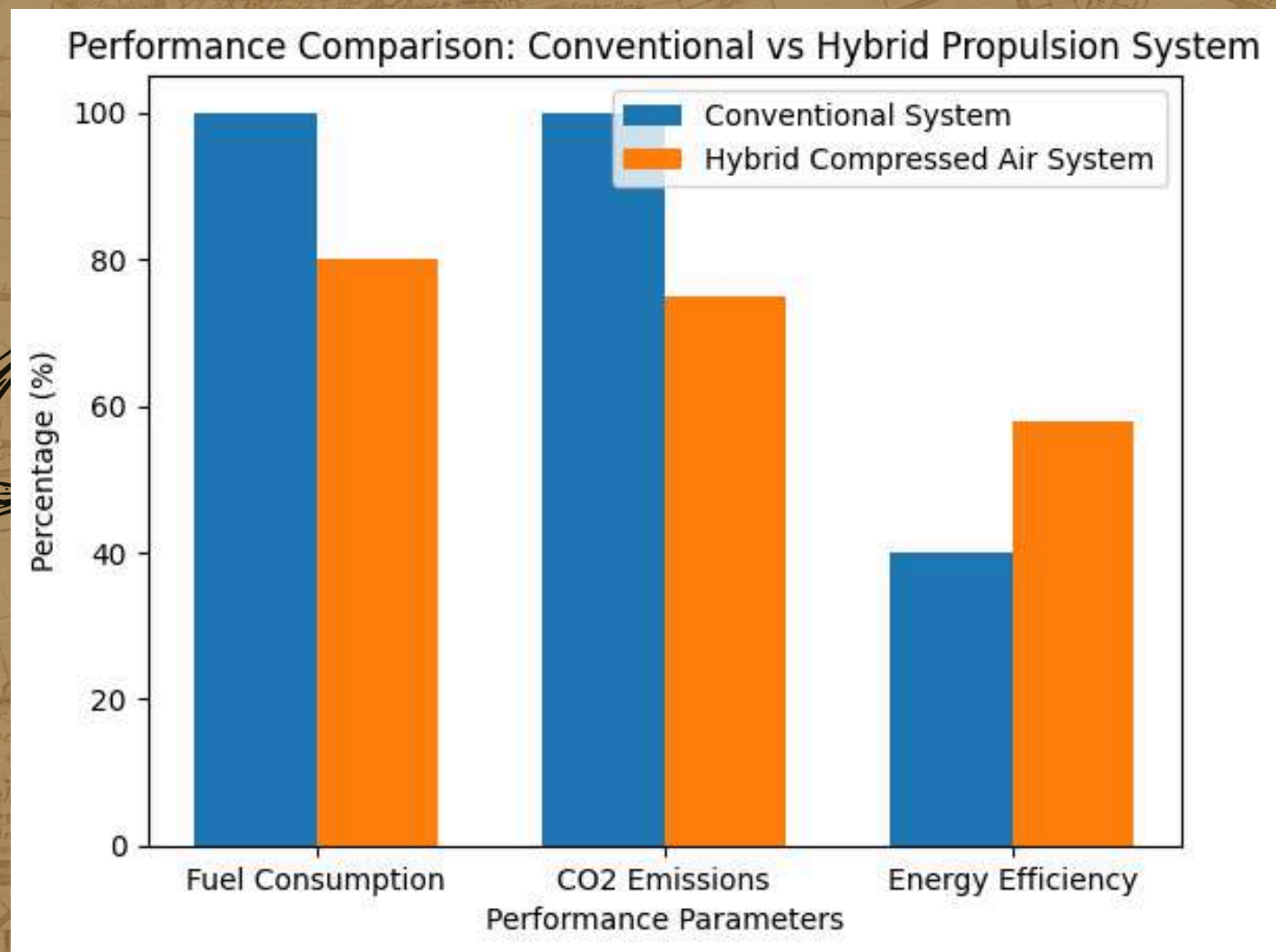
1. Air enters compression chamber.
2. High-pressure air stored.
3. Released compressed air drives secondary turbine.
4. Turbine assists propeller rotation.

Proposed Methodology

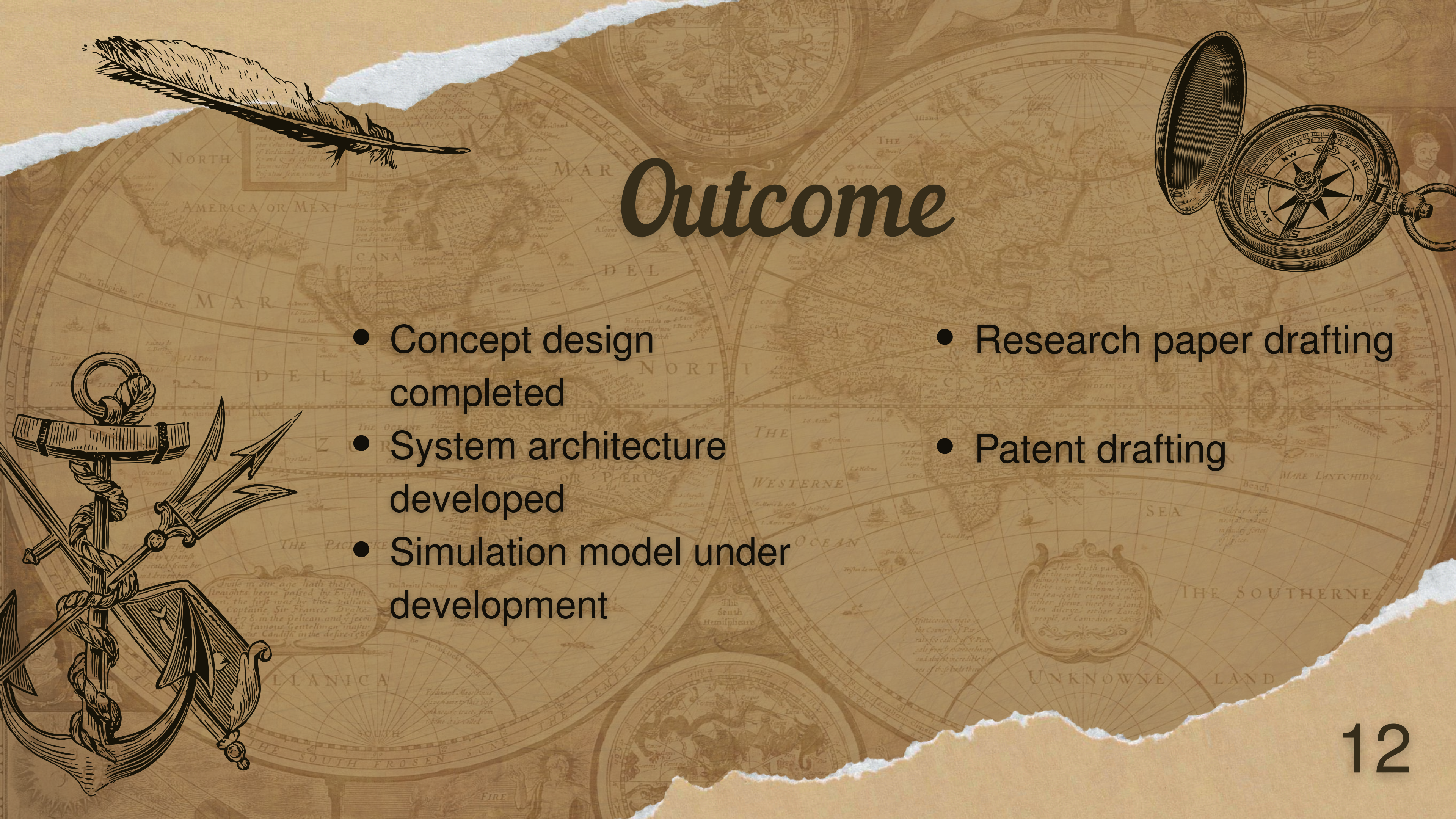
Hybrid Compressed Air Assisted Propulsion System



Results

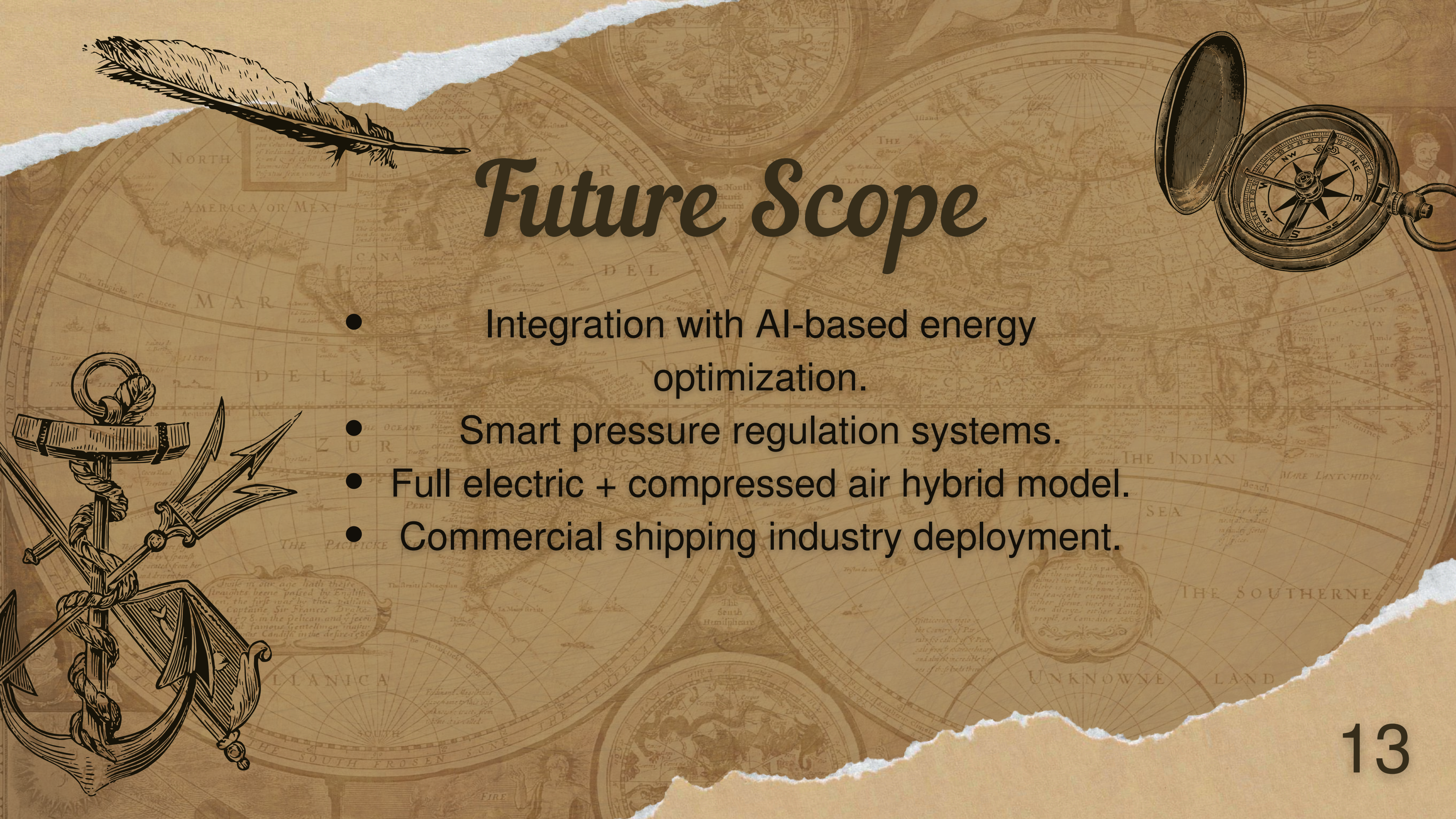


Parameter	Conventional System	Hybrid Compressed Air System
Fuel Consumption	100% Diesel Usage	75–85% Diesel Usage
CO ₂ Emissions	High Emissions	Reduced by 15–25%
Operational Cost	High Fuel Cost	Lower Long-Term Cost
Energy Efficiency	~40%	~55–60%
Auxiliary Energy Recovery	Not Available	Regenerative Airflow Loop



Outcome

- Concept design completed
- System architecture developed
- Simulation model under development
- Research paper drafting
- Patent drafting



Future Scope

- Integration with AI-based energy optimization.
- Smart pressure regulation systems.
- Full electric + compressed air hybrid model.
- Commercial shipping industry deployment.

References

- International Maritime Organization Reports
- Research papers on Compressed Air Energy Storage (CAES)
- Marine Hybrid Propulsion Journals
- IEEE Xplore maritime propulsion studies



Thank You

Driving Maritime Sustainability Through
Innovation